Name of the Student…… **Scholar Number……**

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EXAMINATION: MID TERM **MONTH and YEAR: Nov 2025**

Course: M. Tech **Semester:** I **Branch: CN/AI**

Subject Code: CN-1103/AI-1163 **Subject Name: Wireless Networks**

Maximum Marks: 20 marks

Duration: 60 minutes **Date:** 06-11-2024 **Time:** 09:30 – 10:30 am

Note: 1. Attempt all questions, **2.** Assume necessary data if required and mention it clearly.

Q. No.	Questions	Marks
1	Discuss the evolution of wireless communication technologies from 1G to 6G, focusing on the key enabling technologies, data rate enhancements, and modulation/multiple access techniques in each generation. How have these developments transformed wireless connectivity from voice-only communication to ultra-reliable, low-latency, AI-driven systems? The evolution of cellular networks from the first generation (1G) to the projected sixth generation (6G) has been driven by the need to increase capacity, enhance data rates, improve spectral efficiency, and support increasingly complex multimedia and data-centric services. Each generation is defined by major breakthroughs in multiple access schemes, core network architecture, and performance goals. 1. **First Generation (1G): Analog Systems** (Early 1980s) The first generation of cellular systems were **analog systems**. Feature — Description and Milestones **Access Technology** — **Analog Frequency Division Multiple Access (FDMA)**. These systems were also called analog FDMA systems. **Primary Goal** — To provide **voice communication**. They were designed to carry narrow-band circuit-switched voice services. **2. Second Generation (2G)**: Digital Systems (Early 1990s) The 2G systems were **digital systems**. They were developed in the 1980s and early 1990s Feature — Description and Milestones **Access Technology** — Based on digital technologies, mainly **TDMA (Time Division Multiple Access)** or **CDMA (Code Division Multiple Access)**. Digital voice systems are classified as 2G.	**(8)**

Primary Goal	Primarily designed to improve voice quality and provide a set of rich voice features. They supported low rate data services (16–32 kbps)	
Key Standards	**Global System for Mobile Communications (GSM)**: The world's first digital cellular system, deployed in Germany in **1991/1992**. GSM is a TDMA system. **IS-95 (CDMA)**: A digital cellular system planned for deployment in 1995 in the U.S. based on CDMA technology, which proved capable of a dramatic increase in capacity compared to analog systems.	
Interference/Handoff	Digital systems generally require less Signal-to-Interference Ratio (SIR) (e.g., less than 10 dB), enabling smaller reuse distances and increasing capacity. GSM typically requires only 1 or 2 seconds for handoff once the decision is made, compared to 10 seconds for 1G systems.	

2.5 G General Packet Radio Service (GPRS) – 115.2 kbps
Enhanced Data Rates for GSM Evolution (EDGE) - 500 kbps

3. Third Generation (3G): Digital Voice/Data Systems (Early 2000s)
3G systems were designed to handle both voice and high-speed data.

Feature	Description and Milestones
Access Technology	Primarily based on **CDMA technology**. CDMA systems with packet/circuit switching are classified as 3G systems.
Primary Goal	To support **multimedia services** (including voice, data, and video) and provide higher data rate capabilities.
Key Standards (IMT-2000)	Defined by the ITU as **IMT-2000**. Standards include: **Universal Mobile Telecommunications System (UMTS)** (WCDMA/UTRA-FDD, developed by 3GPP). **cdma2000** (IMT-MC, developed by 3GPP2). **UTRA-TDD/TD-SCDMA** (IMT-TC).
Data Rate Requirements	ITU requirements: **144 kbps** in vehicular environment, **384 kbps** in pedestrian environment, and **2 Mbps** in fixed indoor/pico cell environments.
Evolution/Enhancements (3.5G)	**High-Speed Packet Access (HSPA):** Upgraded UMTS networks, often called 3.5G. Includes **HSDPA** (High Speed Downlink Packet Access, 3GPP Release 5), capable of providing data rates up to **10 Mbps**, and **HSUPA** (High Speed Uplink Packet Access, 3GPP Release 6). **cdma2000 1X EV-DO** (Data Only) and **1X EV-DV** (Data and

Voice). EV-DO achieved burst rates up to **2.4 Mbps**.

4. Fourth Generation (4G): Broadband Digital Systems (Early 2010s)
4G systems represent high-speed, multimedia, and high data rate wireless access.

Feature	Description and Milestones
Access Technology	Use advanced technologies, often based on **OFDM/OFDMA** (Orthogonal Frequency Division Multiplexing/Multiple Access) and advanced single-carrier techniques like **SC-FDMA** (Single-Carrier Frequency Division Multiple Access). CDMA and OFDMA were contrasted as potential solutions for 4G.
Core Architecture	**All Internet Protocol (IP)** packet-switched network. The architecture moves toward a distributed architecture with end-to-end IP.
IMT-Advanced Requirements	**1 Gbps** transmission data rate at a standstill condition. **100 Mbps** transmission data rate at a moving condition. The system is required to have flexible channel bandwidths (5 to 20 MHz, optionally up to 40 MHz).
Key Standards	**Long-Term Evolution (LTE):** Standardized by 3GPP. LTE uses **OFDMA on the downlink** and **SC-FDMA on the uplink**. LTE Release 8 aimed for **100 Mb/s downlink** and **50 Mb/s uplink** peak rates within 20 MHz bandwidth. **LTE-Advanced (LTE-A):** The only truly 4G technology, as defined by 3GPP Releases 10 and 11, capable of reaching 1 Gbit/s peak downlink data rate (with 8x8 MIMO and 100 MHz bandwidth). **Mobile WiMAX (IEEE 802.16e):** Uses OFDMA technology and is often branded as 4G.
Key Technologies	Extensive use of **Multiple-Input Multiple-Output (MIMO)** antenna systems, often combined with OFDMA, to achieve enormous capacity increases and spectral efficiency. Other technologies include **Adaptive Modulation and Coding (AMC)** and **dynamic channel allocation**.

5. Fifth Generation (5G) Cellular (Under Deployment/Research)
5G systems are currently under research and development and are expected to be deployed between 2020 and 2030.

Feature	Description and Milestones
Primary Goal	Massive connectivity and massive capacity. The demanding objectives include 100–1000-fold gains in network sum capacity, connections for at least **100 billion devices**, and reduction of energy-per-bit usage by a factor of 10–100.
Performance Requirements	Offer **10 Gbps peak data rates** with **sub-millisecond latencies and response times**. The performance metrics include providing **100 Mbps in high mobility** and **1 Gbps in stationary/low mobility** applications.
Key Technologies	**Ultra dense networks** using hierarchical cellular architectures. Operation at **higher frequencies**, particularly **mm-wave frequencies**. Extensive use of **Massive MIMO** technology.

	Applications Expected to support low-data-rate machine-to-machine communication, mobile industrial automation, vehicular connectivity, and automated driving (IoT applications). 6. Sixth Generation (6G) and Beyond (Future Vision) Although 6G is not detailed extensively in the provided sources, the trajectory of future systems continues the trends seen in 4G and 5G: • **Future Systems:** Future systems will be based on user demands and will be IP-based. The huge gap between the capabilities of current systems and the vision for future systems indicates that much research and development in wireless communications remains to be done. • **Key Directions:** Ongoing research aims for ubiquitous connectivity, communication, and content. This involves continued evolution towards high data rates and capacity, low latency, increased spectral efficiency, and seamless mobility, leveraging technologies like **OFDMA** and **MIMO**. Future enhancements will focus on integrating different wireless networks, enabling seamless handoff between systems like Wi-Fi and cellular.	
2	A transmitter radiates 20 W of power at a carrier frequency of 2.4 GHz through a unity-gain antenna. (a) Express the transmit power in dBm and dBW. (b) Compute the received power (in dBm) at a distance of 200 m and 10 km under free-space conditions. How much is the difference in dBm when the distance changes? Transmit power: $P_t = 20$ W Frequency: $f = 2.4$ GHz $= 2.4 \times 10^9$ Hz Distance: $d = 200$ m and 10 km $= 10,000$ m Antenna gains: $G_t = G_r = 1$ (unity gain) Speed of light: $c = 3 \times 10^8$ m/s **(a) Express transmit power in dBm and dBW** **In dBW:** $$P_t(\text{dBW}) = 10\log_{10}(P_t)$$ $$P_t(\text{dBW}) = 10\log_{10}(20) = 13.01 \text{ dBW}$$ **In dBm:** $$P_t(\text{dBm}) = P_t(\text{dBW}) + 30 = 13.01 + 30 = 43.01 \text{ dBm}$$ ✅ **Answer (a):** $P_t = 13.01$ dBW $= 43.01$ dBm	**(8)**

	(b) Compute the received power under free-space conditions Use **Friis free-space transmission formula:** $$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$ where $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{2.4 \times 10^9} = 0.125 \text{ m}$ So: $$P_r = 20 \times (1) \times (1) \times \left(\frac{0.125}{4\pi d}\right)^2$$ **(b) Received power using Friis** Compute the factor $\left(\frac{\lambda}{4\pi d}\right)^2$ and then P_r. **At** $d = 200$ m: $$\frac{\lambda}{4\pi d} = \frac{0.125}{4\pi \times 200} \approx 4.9737 \times 10^{-5}$$ $$\left(\frac{\lambda}{4\pi d}\right)^2 \approx 2.4737 \times 10^{-9}$$ $$P_r = 20 \times 2.4737 \times 10^{-9} \approx 4.9474 \times 10^{-8} \text{ W}$$ Convert to dBm: $$P_r(\text{dBm}) = 10 \log_{10}(4.9474 \times 10^{-8} \text{ W}/10^{-3}) \approx -43.06 \text{ dBm}$$ **At** $d = 10{,}000$ m **(10 km):** $$\frac{\lambda}{4\pi d} = \frac{0.125}{4\pi \times 10000} \approx 9.9474 \times 10^{-7}$$ $$\left(\frac{\lambda}{4\pi d}\right)^2 \approx 9.8950 \times 10^{-13}$$ $$P_r = 20 \times 9.8950 \times 10^{-13} \approx 1.9790 \times 10^{-11} \text{ W}$$ Convert to dBm: $$P_r(\text{dBm}) = 10 \log_{10}(1.9790 \times 10^{-11}/10^{-3}) \approx -77.04 \text{ dBm}$$ The difference is 34 dB	
3	In a cellular system, the network operator has chosen a cluster size of N = 7 to balance capacity and coverage. If the system operates in an environment with a path loss exponent of **n = 4**, and we consider the six nearest co-channel interferers: (a) Calculate the co-channel reuse ratio. (b) Determine the Signal-to-Interference Ratio (SIR) in dB at the cell edge (worst-case scenario). (c) Based on your result in (b), would this cluster size be acceptable if the minimum required (SIR) is 18 dB?	**(8)**

🔍 Given Parameters

- **Cluster Size (N):** 7
- **Path Loss Exponent (n):** 4
- **Number of Co-channel Interferers (I_0):** 6 (first tier)

(a) Calculate the Co-channel Reuse Ratio ($Q = D/R$)

The Co-channel Reuse Ratio (Q) relates the distance between co-channel cells (D) to the cell radius (R). It is directly related to the cluster size (N) by the formula:

$$Q = \frac{D}{R} = \sqrt{3N}$$

Substituting the given cluster size ($N = 7$):

$$Q = \sqrt{3 \times 7} = \sqrt{21}$$

$$\boldsymbol{Q \approx 4.5826}$$

(b) Determine the Signal-to-Interference Ratio (SIR) in dB

The worst-case SIR (at the cell edge, where $R = 1$ for the desired signal) is approximated by considering the interference from the first tier of I_0 co-channel cells.

The SIR is calculated using the following relationship, where D is the co-channel distance and n is the path loss exponent:

$$\frac{S}{I} \approx \frac{1}{I_0 \times \left(\frac{D}{R}\right)^{-n}}$$

Since $Q = D/R$, we can rewrite the formula:

$$\frac{S}{I} \approx \frac{Q^n}{I_0}$$

Substituting the values: $Q = \sqrt{21} \approx 4.5826$, $n = 4$, and $I_0 = 6$:

$$\frac{S}{I} \approx \frac{(\sqrt{21})^4}{6} = \frac{(21)^2}{6} = \frac{441}{6}$$

$$\frac{S}{I} = 73.5$$

Now, convert the SIR from a ratio to decibels (dB):

$$\mathrm{SIR}_{\mathrm{dB}} = 10 \log_{10}\left(\frac{S}{I}\right)$$

$$\mathrm{SIR}_{\mathrm{dB}} = 10 \log_{10}(73.5)$$

$$\mathrm{SIR}_{\mathrm{dB}} \approx \mathbf{18.66}\ \mathrm{dB}$$

(c) Acceptability of the Cluster Size

The minimum required SIR for the system is **18 dB**.

- **Calculated** SIR: $18.66\ \mathrm{dB}$
- **Required** SIR: $18.00\ \mathrm{dB}$

Since the calculated worst-case SIR ($18.66\ \mathrm{dB}$) is **greater than** the minimum required SIR ($18.00\ \mathrm{dB}$), the cluster size of $\boldsymbol{N = 7}$ is **acceptable** for this system and radio environment. It provides a slight margin of safety against interference.

4.	Explain the concept of traffic engineering in cellular systems. Cellular network involves critical trade-offs between cost and service quality. Explain the economic and quality trade-off in cellular traffic engineering, using Grade of Service **(GoS)**	**(8)**

and total system capacity as the key metrics.

1. The Economic and Quality Trade-off

This trade-off is the core decision faced by a mobile network operator (MNO) when their network.

Component	Description	Impact on the Trade-off
Grade of Service (GoS)	The probability of a call being blocked or delayed (e.g., $0.01 = 1\%$ blocking). **This is the quality metric.**	**Higher Quality (Lower GoS more channels** and **higher** leading to **higher cost**.
Total System Capacity	The maximum traffic (in Erlangs) the network can handle, which is directly tied to the number of physical channels (C) deployed per cell.	**Higher Capacity:** Requires **equipment** (transceivers, sp licenses), leading to **higher (Capital Expenditure)**.
Economic Cost	The investment in infrastructure (CapEx) to deploy channels and the ongoing operational costs (OpEx).	**Lower Cost:** Achieved by de **fewer channels**, which resu GoS **(higher blocking)** and

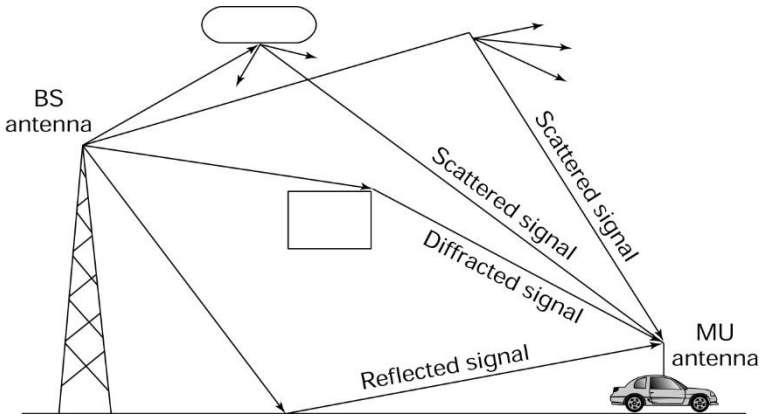
5. Write short notes of two:- **(8)**

(i) Mobile Radio propagation

Mobile radio propagation is the transmission of information via electromagnetic waves through the atmosphere rather than via cables. The core challenge in this field is the **hostile wireless channel,** which causes attenuation, delay, and distortion. The study of mobile radio propagation focuses on terrestrial digital wireless communication systems.

Key Phenomena and Signal Behavior

A common misconception is that a radio frequency (RF) signal travels from an antenna via a single direct path (line of sight) to a receiver. In reality, **multiple copies** of the signal may reach the receiver, a phenomenon known as **multipath**. The most general case of signal reception consists of a direct path, a reflected path, a scattered path, and a diffracted path.



Wave propagation involves several different behaviors:

- **Reflection:** A signal bounces back when hitting objects that are very large and smooth, such as walls, buildings, the surface of the earth, or metal objects.
- **Diffraction:** Occurs when an object with rough surfaces is in the path of the RF signal, causing the signal to bend. Diffraction loss is often severe in urban areas with high-rise buildings.
- **Scattering:** Caused by small objects like foliage, rocks, and sand, or airborne substances such as rain or heavy dust.
- **Absorption:** Materials like concrete, wood, and asphalt absorb the RF signal.
- **Refraction:** The bending of the signal as it moves from one medium to another of a different density.

Signal Loss and Fading

The behavior of the RF signal impacts the speed of transmission and the distance achieved between devices. A critical impact is **attenuation**, or the loss of signal strength.

Fading is the variation in signal amplitude at the receiver caused by characteristics and changes in the signal path. It is generally caused by variation in distance, changes in environmental characteristics, the presence of multiple signal paths, and relative motion between the transmitter and receiver.

Major types of fading include:

1. **Free Space Path Loss (FSPL):** The natural loss of signal strength as it travels through space, irrespective of environmental objects. Received power decreases with the **square of the distance** (inverse-square law) in an ideal environment.
2. **Shadow Fading (Long-term fading):** Occurs due to obstacles like **large hills or buildings** obstructing the signal path, causing the received signal strength to fluctuate around an average value.
3. **Multipath Fading (Rayleigh Fading/Short-term fading):** Occurs because reflected signals travel longer distances, causing them to be delayed and resulting in a phase shift. This often leads to the reflected signals canceling out the direct path signal, causing a weaker overall signal. In small-scale fading, the received power can vary by 30 or 40 dB over a fraction of a wavelength.
4. **Doppler Shift:** A type of fading introduced by rapid movement of either the transmitter or the receiver towards or away from each other, which causes a signal frequency change.

Propagation Models

Propagation models are used to predict the average received signal strength.

1. **Large-Scale Propagation Models:** These models characterize signal strength over large separation distances (hundreds or thousands of meters) and predict the mean signal strength to estimate radio coverage area.
    - The **Free Space Propagation Model** is a basic example, used when there is an unobstructed line-of-sight path. It employs the **Friis free space equation** to relate transmitted power, antenna gains, and received power.
    - **Simplified power falloff models** are good for high-level analysis, often capturing path loss using the path loss exponent (γ), which is determined empirically.
2. **Small-Scale or Fading Models:** These models characterize the rapid fluctuations of the received signal strength over very short travel distances (a few wavelengths) or short time durations.
3. **Ray Tracing Models:** These models simplify Maxwell's equations by representing wavefronts as simple particles. They require detailed site parameters, including geometry and dielectric properties, and can model reflections, scattering, and diffraction.
    - The **Two Ray Model** specifically considers one LOS path and one ground bounce, predicting power fall-off proportional to d^4 at distances greater than a critical distance ($d>d_c$).
4. **Empirical Models:** These models, such as the **Okumura model** and its analytical approximation, the **Hata model**, are based on field data and are commonly used in cellular system simulations.

The study of mobile radio propagation is analogous to **understanding the acoustics of a concert hall**, where the same sound emitted from the stage is drastically altered by reflections, absorption, and interference patterns from walls and objects before it reaches different listeners, resulting in varying sound quality and intensity depending on the listener's precise location.

(ii) Media-access techniques

Medium Access Control (MAC) is essential for managing communications on a shared channel, determining **who transmits data**, and establishing the **timing and method of data transmission**. Its primary goal is to ensure the **efficient and fair use of communication resources** by handling multiple stations on a single link.

The main techniques for sharing a single link (multiplexing) often include FDMA, TDMA, and CDMA, which are commonly used in circuit-switched voice networks:

- **Frequency Division Multiple Access (FDMA):** This method divides the bandwidth into **separate (unique) carrier frequency bands** for each user. FDMA was utilized by all **1G (first-generation) systems**.
- **Time Division Multiple Access (TDMA):** This technique allocates a **separate (unique) time slot** to each user. It utilizes the same carrier frequency, which is efficiently split into time slots. Most **2G systems** employ TDMA.
- **Code Division Multiple Access (CDMA):** CDMA allows different users to use the **same frequency simultaneously** by assigning a **separate (unique) orthogonal code** to each user. CDMA is considered **conflict-free**, unlike contention-based access methods. Some 2G systems and most **3G systems** use CDMA. CDMA can be implemented using Direct Sequence (DS) or Frequency Hopping (FH) methodologies.

Another important access technique, used primarily in Ethernet due to its simplicity, is **CSMA/CD (Carrier Sense Multiple Access with Collision Detection)**. In this contention-based protocol, stations **listen before transmitting** to prevent collisions. If a collision is detected, transmission is delayed and retried.

(iii) Channel Assignment.

Channel Assignment Strategy

Channel assignment strategies in mobile communications determine how radio frequencies are distributed among cells within a network to effectively manage calls.

There are three main types of channel assignment approaches:

1. **Static/Fixed Channel Assignment:** This strategy involves the pre-allocation of specific channels to each cell. While it provides simplicity, its major limitation is the limited flexibility it offers when network conditions change.
2. **Dynamic Channel Assignment:** This method allocates channels on demand from a central pool in real time. Dynamic assignment is beneficial because it adapts to network conditions and improves efficiency by minimizing interference.
3. **Hybrid Channel Approaches:** These approaches combine the characteristics of both static and dynamic methods. The purpose of hybrid strategies is to leverage the advantages of both to optimize channel utilization and overall network performance.

**********END**********